FLEET TEAM DIGEST

737MAX-FTD-34-24010

Issue Title : DRV-2120: Optical Encoder Disk Reliability

Airplane Model	ATA	Minor Model(s)
737MAX	3441-11	-8, -8200, -9, -BBJ

Other Model(s)	ECCN
787	9E991

Originated Date	Last Revised Date	Created On
10/25/2024	04/07/2026	03/17/2026

Estimated Completion Date	Next Update Date
11/30/2026	06/17/2026

Revision Description

Updated status and Milestones.

Applicability

All 737-8/8200/9/BBJ models with Collins WXR-2100A Weather Radar (WXR) Antenna Drive Unit DRV-21X0 installed ('X' is variable) (Part Number 822-2131-003/822-2130-001).

Description

On the 737-8/8200/9/BBJ model airplanes, there are reports of reduced reliability of the WXR Antenna Drive Unit DRV-21X0 ('X' is variable) performance due to an optical encoder degradation issue. This condition results in a WXR System fault (e.g., 'WXR FAIL) being displayed to the Flight Crew.

The following maintenance messages are indicative of Antenna Drive Unit DRV-21X0 ('X' is variable) faults due to an issue with the optical encoder disk:

-34-43101 - “WXR Antenna Drive is Failed” Detected By: WXR PROCESSOR 1

-34-43201 - “WXR Antenna Drive is Failed” Detected By: WXR PROCESSOR 2

Note: Weather detection / display may be degraded while the WXR fault is active. For hard, persistent failures, depending on how far off the tilt and scan values are, the display could show weather at incorrect locations. For momentary or intermittent faults, there may be no degradation in weather detection or display and the faults are normally only seen in Maintenance Messages after the flight.

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Status

Collins Aerospace and Boeing have analyzed shop repair data for returned Antenna Drive Units and determined that the root cause is due to optical encoder disk emulsion crazing and flaking off. Boeing has initiated an SRP (787-SRP-34-0182 WXR Antenna Optical Disk Reliability) to manage, support, and coordinate the development efforts of a new encoder disk. Corrective action from the 787 investigation will be applicable to 737 MAX since they use the same antenna drive technology.

Magnetic Encoder Technology Solution Selection Path Background:

The flight test campaign to study the radome environment and robustness testing of new magnetic encoder technology was completed in Sept-2021.

A Non-Advocate Review (NAR), consisting of an enterprise wide panel of Boeing subject matter experts and Collins engineers was held on 18-Oct-2021 to evaluate the test and analysis findings, and to recommend a technical solution. After a thorough investigation, it was mutually decided to push forward with magnetic encoder technology as a replacement for the current optical encoder disk.

The magnetic encoder technology that was selected as the solution was to be interchangeable with the existing optical encoder disk with no software changes required. Boeing and Collins proceeded with the change committal before moving to a production incorporation and supplier service bulletin release.

In addition to the normal operational and requirements testing, Collins also used the new magnetic encoder disk during WXR V2.0 ThreatTrack flight test to build more confidence in the improved reliability.

In late 2023, Collins conducted additional variation temperature testing on the proposed DRV-2120 magnetic encoder fix. After extensive testing and validation, Collins determined that the anticipated solution was no longer technically viable.

Future (TBD) Technology Solution Selection Path:

Upon determination that the magnetic encoder technology solution selection path was no longer technically viable, Collins engineering immediately launched an activity to identify alternative solutions to improve the performance of the weather radar system.

Collins is currently working on a potential solution of a parylene coated optical encoder disk. Collins completed several functional tests using multiple encoder sample sizes to determine parylene effectiveness and durability including: Optimal Coating Validation, High Level Functional Test, System Test Using Encapsulated Discs, and Parylene Durability test.

A joint Boeing/Collins meeting to discuss the development test results and recommended path forward is expected to be completed by November 2024.

Collins is currently working on the potential solution of applying a parylene coating to the optical encoder disk:

- Solution initially yielded positive results during extensive system and reliability testing.
- Parylene proved to provide strong resistance to moisture and offered promising potential.

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- Parylene seals the surface and provides a barrier against moisture.
- Extended humidity, temperature and pressure testing which identified susceptibility to moisture intrusion.
- Destructive testing identified abnormalities in geometry of disk creates gaps in material that Parylene is not designed to fill.
- Potential sealants have been identified which could be used to fill the gaps.
- Testing of the sealants along with the Parylene coating has yielded positive results and testing is currently in progress.

Interim Action

Should operators experience a high rate of Antenna Drive failures associated with MMs 34-43101 and 34-43201, please contact Collins Customer Support Engineering to assist with the process of proactive removal and replacement of encoder disk based on flight hours and flight cycles.

Final Action

When available, the Final Action will be to incorporate the equipment upgrade via Supplier Service Bulletin.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: March 2020 (Complete)

Magnetic Encoder Technology:

Solution Selected: October 2021 (Complete)

Change Committed: August 2023 (Complete)

Solution Determined Not Technically Viable: January 2024 (Complete)

Future (TBD) Technology:

Joint Technical Review (Collins & Boeing): September 2024 (Complete)

Parylene Viability Decision Point: July 2025 (Complete)

Solution Identified: August 2025 (Complete)

Sealant Selected: April 2026 (Target)

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Change Committed: August 2026 (Target)

Production Incorporation: September 2026 (Target)

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Related Problem (SRP)	787-SRP-34-0182		
Other	CN-AA90076		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Antenna Drive Unit (DRV-2120)	PN 822-2131-003	Collins Aerospace	DRV-2120
Antenna Drive Unit (DRV-2110)	PN 822-2130-001	Collins Aerospace	DRV-2110

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Flight Deck Effects and Maintenance Messages : Issue that occurs frequently and causes a flight deck effect or maintenance message

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